

Python Practice Workbook - Expanded Edition

Conditionals, Loops, Loop Control, Number Logic and Patterns

65 Questions | Two Test Cases Each | Progressive Difficulty | Mixed Input Styles

Instructions

- Read the complete question before coding and identify which construct is being tested.
- Use the programming construct specified in the Practice Rule; avoid shortcut functions when prohibited.
- Match required output text, capitalization, spacing, and number formatting exactly.
- Every question has two visible test cases. Also create at least two of your own hidden test cases.
- Input style is intentionally varied. Some questions use one value per line, some use multiple values on one line, and some explicitly require prompt text inside input().
- When a question says to use a prompt such as input("Enter number: "), include that prompt in your code even though the test-case boxes below show only the values supplied by the user.

1. Basic Input, Output, Variables and Arithmetic

Question 1: Shopping Bill

Difficulty: Core

Write a Python program that reads the quantity of an item and its price per item. Calculate and print the total amount to be paid.

Input Format: Two lines: an integer quantity q, then an integer price p.

Output Format: Print the total amount as an integer.

Practice Rule: Use variables, input(), arithmetic operators, and print().

Test Case 1

Input values

```
3
250
```

Expected Output

```
750
```

Test Case 2

Input values

```
7
99
```

Expected Output

693

Question 2: Minutes to Hours and Minutes

Difficulty: Core

Write a program that reads a total number of minutes and converts it into complete hours and remaining minutes.

Input Format: One integer m representing total minutes.

Output Format: Print hours and remaining minutes in the format shown.

Practice Rule: Use integer division and modulus.

Test Case 1

Input values

135

Expected Output

```
hours: 2  
minutes: 15
```

Test Case 2

Input values

59

Expected Output

```
hours: 0  
minutes: 59
```

Question 3: Rectangle Measurements

Difficulty: Core

Read the length and breadth of a rectangle. Print its area and perimeter.

Input Format: Two integers l and b on separate lines.

Output Format: Print area on the first line and perimeter on the second line.

Practice Rule: Use only basic arithmetic expressions.

Test Case 1

Input values

```
8
5
```

Expected Output

```
40
26
```

Test Case 2

Input values

```
12
10
```

Expected Output

```
120
44
```

Question 4: Three-Part Expense Split

Difficulty: Moderate

A student has a fixed budget. Read the total budget and the amounts spent on food and travel. Calculate the remaining amount. If the spending is greater than the budget, the result will be negative; do not apply any condition yet.

Input Format: Read three integers on one line: budget food travel.

Output Format: Print remaining: value.

Practice Rule: Read all three values from a single input line using split().

Test Case 1

Input values

```
5000 1250 900
```

Expected Output

```
remaining: 2850
```

Test Case 2

Input values

```
2000 1500 800
```

Expected Output

```
remaining: -300
```

Question 5: Prompted Percentage Calculator

Difficulty: Moderate

Read obtained marks and maximum marks, then calculate the percentage. This question is also meant to practice placing a message inside input().

Input Format: Use the prompts Enter marks: and Enter maximum: while reading two floating-point values on separate lines.

Output Format: Print percentage: followed by the value rounded to two decimals.

Practice Rule: Your input statements must contain prompt strings, for example input("Enter marks: "). Use arithmetic and formatted output.

Test Case 1

Input values

```
375  
500
```

Expected Output

```
percentage: 75.00
```

Test Case 2

Input values

```
47.5  
50
```

Expected Output

```
percentage: 95.00
```

2. Conditional Statements: if, elif and else

Question 6: Age Category

Difficulty: Core

Read a person's age and classify it as Child, Teen, Adult, or Senior. Use: below 13 = Child, 13-19 = Teen, 20-59 = Adult, 60 and above = Senior.

Input Format: One integer age.

Output Format: Print exactly one category name.

Practice Rule: Use an if-elif-else ladder.

Test Case 1

Input values

12

Expected Output

Child

Test Case 2

Input values

67

Expected Output

Senior

Question 7: Temperature Status

Difficulty: Core

Read an integer temperature in degree Celsius. Print Cold if it is below 15, Pleasant if it is from 15 to 25 inclusive, Warm if it is from 26 to 35 inclusive, and Hot if it is above 35.

Input Format: One integer temperature.

Output Format: Print exactly one status word.

Practice Rule: Use ordered conditions carefully.

Test Case 1

Input values

22

Expected Output

Pleasant

Test Case 2

Input values

39

Expected Output

Hot

Question 8: Largest of Three Numbers

Difficulty: Core

Read three integers and print the largest value. The values may be equal.

Input Format: Three integers a, b, and c on separate lines.

Output Format: Print the largest integer.

Practice Rule: Solve using conditional statements; do not use max().

Test Case 1

Input values

18
42
31

Expected Output

42

Test Case 2

Input values

25
25
10

Expected Output

25

Question 9: Electricity Bill Slab

Difficulty: Challenge

Read electricity units consumed. Calculate the bill using these slabs: first 100 units at Rs. 2 per unit, next 100 units at Rs. 3 per unit, and every unit above 200 at Rs. 5 per unit. The slabs are cumulative.

Input Format: One non-negative integer u.

Output Format: Print bill: followed by the integer amount.

Practice Rule: Use if-elif-else and calculate cumulative slab charges carefully.

Test Case 1

Input values

80

Expected Output

bill: 160

Test Case 2

Input values

250

Expected Output

bill: 750

Question 10: Login Attempt Decision

Difficulty: Moderate

Read a username and a four-digit PIN. If username is admin and PIN is 2468, print Access Granted. If the username is correct but the PIN is wrong, print Wrong PIN. Otherwise print Unknown User.

Input Format: First line: username string. Second line: integer PIN. Read the username using input("Username: ") and the PIN using input("PIN: ").

Output Format: Print exactly one decision message.

Practice Rule: Use nested or ordered conditional statements. This is a prompted-input practice question.

Test Case 1

Input values

admin
2468

Expected Output

Access Granted

Test Case 2

Input values

admin
1111

Expected Output

Wrong PIN

3. match-case Statement

Question 11: Day Number to Day Name

Difficulty: Core

Read a number from 1 to 7 and print the corresponding day name where 1 is Monday and 7 is Sunday. For any other value, print Invalid.

Input Format: One integer d.

Output Format: Print the day name or Invalid.

Practice Rule: Use match-case.

Test Case 1

Input values

3

Expected Output

Wednesday

Test Case 2

Input values

9

Expected Output

Invalid

Question 12: Simple Calculator

Difficulty: Core

Read two integers followed by an operator (+, -, *, or /). Perform the requested operation. For division, print the result as a floating-point value. If the operator is unsupported, print Invalid operator.

Input Format: Three lines: integer a, integer b, and operator op.

Output Format: Print the calculated result or Invalid operator.

Practice Rule: Use match-case on the operator.

Test Case 1

Input values

```
12
4
*
```

Expected Output

```
48
```

Test Case 2

Input values

```
20
5
/
```

Expected Output

```
4.0
```

Question 13: Grade Converter

Difficulty: Core

Read marks out of 100 and print the grade using this scale: 90 and above = A, 80-89 = B, 70-79 = C, 60-69 = D, below 60 = F.

Input Format: One integer marks.

Output Format: Print A, B, C, D, or F.

Practice Rule: Use match-case with guards.

Test Case 1

Input values

```
86
```

Expected Output

B

Test Case 2

Input values

54

Expected Output

F

Question 14: Traffic Signal Command

Difficulty: Moderate

Read a traffic-light command: RED, YELLOW, or GREEN. Print Stop, Wait, or Go respectively. Any other command must print Invalid signal.

Input Format: One uppercase string command.

Output Format: Print exactly one message.

Practice Rule: Use match-case with string cases and a default case.

Test Case 1

Input values

RED

Expected Output

Stop

Test Case 2

Input values

BLUE

Expected Output

Invalid signal

Question 15: Menu-Based Area Calculator

Difficulty: Challenge

Read a command followed by the required dimensions. If the command is SQUARE, read one side and print its area. If it is RECTANGLE, read length and breadth and print the area. If it is TRIANGLE, read base and height and print $0.5 * \text{base} * \text{height}$. For any other command print Invalid shape.

Input Format: First line is SQUARE, RECTANGLE, TRIANGLE, or another word. Remaining lines contain the dimensions required by that command.

Output Format: Print area: value. Triangle area must be displayed with one decimal place.

Practice Rule: Use match-case. Each case should read only the inputs needed for that shape.

Test Case 1

Input values

```
SQUARE
6
```

Expected Output

```
area: 36
```

Test Case 2

Input values

```
TRIANGLE
10
7
```

Expected Output

```
area: 35.0
```

4. while Loop

Question 16: Sum of Even Numbers

Difficulty: Core

Read n and calculate the sum of all even numbers from 2 up to n. If n is odd, the last added value is n-1.

Input Format: One integer n.

Output Format: Print the sum as sum: value.

Practice Rule: Use a while loop.

Test Case 1

Input values

12

Expected Output

sum: 42

Test Case 2

Input values

9

Expected Output

sum: 20

Question 17: Reverse an Integer

Difficulty: Core

Read a positive integer and print its digits in reverse order as an integer.

Input Format: One positive integer n.

Output Format: Print the reversed integer.

Practice Rule: Use a while loop with % and //; do not convert the number to a string.

Test Case 1

Input values

4821

Expected Output

1284

Test Case 2

Input values

1200

Expected Output

21

Question 18: Count Digits

Difficulty: Core

Read a non-negative integer and count how many digits it contains. The number 0 has one digit.

Input Format: One non-negative integer n.

Output Format: Print the digit count.

Practice Rule: Use a while loop.

Test Case 1

Input values

90715

Expected Output

5

Test Case 2

Input values

0

Expected Output

1

Question 19: Digital Sum Until One Digit

Difficulty: Challenge

Read a positive integer. Repeatedly add its digits until the result becomes a single digit. Print that final digit.
Example: 9875 -> 29 -> 11 -> 2.

Input Format: One positive integer n.

Output Format: Print the final single digit.

Practice Rule: Use nested/repeated while-loop logic. Do not convert the number to a string.

Test Case 1

Input values

9875

Expected Output

2

Test Case 2

Input values

99999

Expected Output

9

Question 20: Prompted PIN Attempts

Difficulty: Challenge

A user gets at most three attempts to enter the PIN 4321. Read attempts one by one. Print Access Granted immediately when the correct PIN is entered. If all three attempts are wrong, print Card Blocked.

Input Format: Read up to three integers. Use input("Enter PIN: ") for each attempt that is actually needed.

Output Format: Print Access Granted or Card Blocked.

Practice Rule: Use a while loop, an attempt counter, and stop asking for input after success.

Test Case 1

Input values

1111
4321

Expected Output

Access Granted

Test Case 2

Input values

```
1000
2000
3000
```

Expected Output

```
Card Blocked
```

5. for Loop Basics

Question 21: Sum a List of Numbers

Difficulty: Core

Read space-separated integers and calculate their sum by visiting each number one by one.

Input Format: One line containing space-separated integers.

Output Format: Print the result in the format sum: value.

Practice Rule: Use a for loop; do not use sum().

Test Case 1

Input values

```
1 10 20 30 40 50
```

Expected Output

```
sum: 151
```

Test Case 2

Input values

```
1 -1 2 -2 3 -4
```

Expected Output

```
sum: -1
```

Question 22: Multiplication Table

Difficulty: Core

Read an integer n and print its multiplication table from 1 to 10. Each line must follow the format $n * i = \text{result}$.

Input Format: One integer n .

Output Format: Print 10 lines of the multiplication table.

Practice Rule: Use a for loop.

Test Case 1

Input values

4

Expected Output

```
4 * 1 = 4
4 * 2 = 8
4 * 3 = 12
4 * 4 = 16
4 * 5 = 20
4 * 6 = 24
4 * 7 = 28
4 * 8 = 32
4 * 9 = 36
4 * 10 = 40
```

Test Case 2

Input values

2

Expected Output

```
2 * 1 = 2
2 * 2 = 4
2 * 3 = 6
2 * 4 = 8
2 * 5 = 10
2 * 6 = 12
2 * 7 = 14
2 * 8 = 16
2 * 9 = 18
2 * 10 = 20
```

Question 23: Count Vowels

Difficulty: Core

Read a single lowercase word and count how many vowels (a, e, i, o, u) it contains.

Input Format: One lowercase word.

Output Format: Print the vowel count.

Practice Rule: Use a for loop to inspect each character.

Test Case 1

Input values

education

Expected Output

5

Test Case 2

Input values

rhythm

Expected Output

0

Question 24: Second Largest Distinct Value

Difficulty: Challenge

Read space-separated integers and find the second largest distinct value. You may assume that at least two distinct values are present.

Input Format: One line of space-separated integers.

Output Format: Print second: value.

Practice Rule: Use a for loop. Do not use sorted(), sort(), max(), or set().

Test Case 1

Input values

4 9 2 9 7 5

Expected Output

second: 7

Test Case 2

Input values

```
10 3 8 6
```

Expected Output

```
second: 8
```

Question 25: Word Length Report

Difficulty: Moderate

Read space-separated words. For every word, print the word followed by its length in the format word:length. At the end, print total: followed by the number of words.

Input Format: One line containing words separated by spaces.

Output Format: Print one report line per word, then the total line.

Practice Rule: Use a for loop and a counter. Do not use len() on the complete list to obtain the total count.

Test Case 1

Input values

```
red blue green
```

Expected Output

```
red:3  
blue:4  
green:5  
total: 3
```

Test Case 2

Input values

```
python loop
```

Expected Output

```
python:6  
loop:4  
total: 2
```

6. range() Function

Question 26: Print Odd Numbers

Difficulty: Core

Read n and print all odd numbers from 1 to n inclusive, one per line.

Input Format: One integer n.

Output Format: Print the required odd numbers.

Practice Rule: Use range() with a suitable start and step.

Test Case 1

Input values

6

Expected Output

1
3
5

Test Case 2

Input values

10

Expected Output

1
3
5
7
9

Question 27: Multiples of Three in a Range

Difficulty: Core

Read two integers a and b and print every number from a to b inclusive that is divisible by 3.

Input Format: Two integers a and b on separate lines.

Output Format: Print each matching number on a new line.

Practice Rule: Use range() in a for loop.

Test Case 1

Input values

```
4
15
```

Expected Output

```
6
9
12
15
```

Test Case 2

Input values

```
18
24
```

Expected Output

```
18
21
24
```

Question 28: Sum of Squares

Difficulty: Core

Read n and calculate $1^2 + 2^2 + \dots + n^2$ using a loop.

Input Format: One positive integer n.

Output Format: Print the sum of squares.

Practice Rule: Use range(); do not use a direct formula.

Test Case 1

Input values

```
4
```

Expected Output

30

Test Case 2

Input values

6

Expected Output

91

Question 29: Descending Multiples

Difficulty: Moderate

Read n. Starting from n and moving downward to 1, print only the numbers divisible by 4.

Input Format: One positive integer n.

Output Format: Print each selected number on a new line in descending order.

Practice Rule: Use range() with a negative step.

Test Case 1

Input values

18

Expected Output

16
12
8
4

Test Case 2

Input values

9

Expected Output

8
4

Question 30: Custom Sequence Generator

Difficulty: Challenge

Read start, stop, and step on one line. Print the sequence generated from start up to but not including stop using the given step. The step will never be zero.

Input Format: Three integers on one line: start stop step.

Output Format: Print generated values on one line separated by one space. Do not print an extra label.

Practice Rule: Use range(start, stop, step). This question practices both positive and negative step values.

Test Case 1

Input values

```
2 12 3
```

Expected Output

```
2 5 8 11
```

Test Case 2

Input values

```
10 1 -3
```

Expected Output

```
10 7 4
```

7. Nested for Loops and Patterns

Question 31: Increasing Star Pattern

Difficulty: Core

Read n and print n rows. Row 1 contains one *, row 2 contains two *, and so on until row n.

Input Format: One integer n.

Output Format: Print the star pattern.

Practice Rule: Use nested for loops.

Test Case 1

Input values

```
4
```

Expected Output

```
*  
**  
***  
****
```

Test Case 2

Input values

```
2
```

Expected Output

```
*  
**
```

Question 32: Increasing Number Pattern

Difficulty: Core

Read n and print n rows. In row i, print the numbers from 1 to i separated by one space.

Input Format: One integer n.

Output Format: Print the number pattern.

Practice Rule: Use nested for loops.

Test Case 1

Input values

```
4
```

Expected Output

```
1  
1 2  
1 2 3  
1 2 3 4
```

Test Case 2

Input values

```
3
```

Expected Output

```
1  
1 2  
1 2 3
```

Question 33: Rectangle of Hash Symbols

Difficulty: Core

Read rows r and columns c . Print a rectangle containing r rows and c hash (#) symbols in each row.

Input Format: Two positive integers r and c .

Output Format: Print the rectangular pattern.

Practice Rule: Use one loop for rows and one inner loop for columns.

Test Case 1

Input values

```
3
5
```

Expected Output

```
#####
#####
#####
```

Test Case 2

Input values

```
2
3
```

Expected Output

```
###
###
```

Question 34: Right-Aligned Star Triangle

Difficulty: Challenge

Read n and print a right-aligned triangle containing n rows. Row 1 has $n-1$ spaces followed by one star; the last row has n stars. Use one space character for each leading position and no spaces between stars.

Input Format: One positive integer n .

Output Format: Print the required pattern.

Practice Rule: Use nested loops; one part should create leading spaces and another should create stars.

Test Case 1

Input values

```
4
```

Expected Output

```
*
**
***
****
```

Test Case 2

Input values

3

Expected Output

```
*
**
***
```

Question 35: Multiplication Grid

Difficulty: Moderate

Read n and print an n by n multiplication grid. In row i and column j , print $i*j$. Separate values by one space.

Input Format: One positive integer n .

Output Format: Print n rows of multiplication values.

Practice Rule: Use two nested for loops. Do not pre-build a multiplication table list.

Test Case 1

Input values

3

Expected Output

```
1 2 3
2 4 6
3 6 9
```

Test Case 2

Input values

2

Expected Output

```
1 2
2 4
```

8. break Statement

Question 36: First Number Divisible by 5

Difficulty: Core

Read n and m. Starting from n, find the first number before m that is divisible by 5. Print it and stop searching. If none exists, print Not found.

Input Format: Two integers n and m.

Output Format: Print the first matching value or Not found.

Practice Rule: Use break when the first match is found.

Test Case 1

Input values

```
1
10
```

Expected Output

```
5
```

Test Case 2

Input values

```
16
19
```

Expected Output

```
Not found
```

Question 37: First Negative Number

Difficulty: Core

Read space-separated integers. Print the first negative number encountered and stop. If the list contains no negative number, print No negative.

Input Format: One line of space-separated integers.

Output Format: Print the first negative number or No negative.

Practice Rule: Use a for loop and break.

Test Case 1

Input values

```
4 8 -3 -9 5
```

Expected Output

-3

Test Case 2

Input values

2 6 10 14

Expected Output

No negative

Question 38: First Common Multiple

Difficulty: Core

Read n and m. Search from n to m inclusive and print the first number divisible by both 4 and 6. If there is no such value, print Not found.

Input Format: Two integers n and m.

Output Format: Print the first common multiple or Not found.

Practice Rule: Stop immediately after the first valid number.

Test Case 1

Input values

10
40

Expected Output

12

Test Case 2

Input values

25
30

Expected Output

Not found

Question 39: Stop at Sentinel

Difficulty: Moderate

Read space-separated integers. Add values from left to right, but stop completely when the value 0 is encountered. Numbers after 0 must be ignored.

Input Format: One line of space-separated integers.

Output Format: Print sum: value.

Practice Rule: Use break when the sentinel value 0 is reached.

Test Case 1

Input values

```
5 7 3 0 100 200
```

Expected Output

```
sum: 15
```

Test Case 2

Input values

```
0 9 8
```

Expected Output

```
sum: 0
```

Question 40: Password Character Check

Difficulty: Moderate

Read a password string and scan it from left to right. Print the first digit found and stop. If no digit exists, print No digit.

Input Format: One string password.

Output Format: Print first digit: d, or No digit.

Practice Rule: Use a loop and break. Do not use regular expressions.

Test Case 1

Input values

```
abc7xy2
```

Expected Output

```
first digit: 7
```

Test Case 2

Input values

securePass

Expected Output

No digit

9. continue Statement

Question 41: Skip Multiples of Three

Difficulty: Core

Read n and print all numbers from 1 to n except numbers divisible by 3.

Input Format: One integer n.

Output Format: Print each accepted number on a new line.

Practice Rule: Use continue to skip multiples of 3.

Test Case 1

Input values

7

Expected Output

1
2
4
5
7

Test Case 2

Input values

4

Expected Output

1
2
4

Question 42: Add Only Positive Values

Difficulty: Core

Read space-separated integers and calculate the sum of only the positive values. Ignore zero and negative values.

Input Format: One line of space-separated integers.

Output Format: Print the positive-value sum.

Practice Rule: Use continue for values that should not be added.

Test Case 1

Input values

```
5 -2 7 0 -1 4
```

Expected Output

```
16
```

Test Case 2

Input values

```
-5 -1 0 8
```

Expected Output

```
8
```

Question 43: Print Only Consonants

Difficulty: Core

Read a lowercase word and print each consonant on a new line. Skip vowels.

Input Format: One lowercase word.

Output Format: Print consonants in their original order.

Practice Rule: Use continue when the current character is a vowel.

Test Case 1

Input values

```
orange
```

Expected Output

```
r  
n  
g
```

Test Case 2

Input values

```
code
```

Expected Output

```
c  
d
```

Question 44: Filtered Sum in a Range

Difficulty: Challenge

Read a and b. Add every integer from a to b inclusive except values divisible by 4 or 6.

Input Format: Two integers a and b on separate lines, where $a \leq b$.

Output Format: Print sum: value.

Practice Rule: Use continue to skip a value whenever it is divisible by 4 or by 6.

Test Case 1

Input values

```
1  
10
```

Expected Output

```
sum: 37
```

Test Case 2

Input values

```
10  
15
```

Expected Output

```
sum: 63
```

Question 45: Clean a Word

Difficulty: Moderate

Read a word containing letters and digits. Print only the letters in their original order on one line; skip all numeric characters.

Input Format: One alphanumeric string.

Output Format: Print the letters-only result.

Practice Rule: Use a for loop and continue. You may use character comparison or `isdigit()` only for deciding what to skip.

Test Case 1

Input values

```
pyt3hon7
```

Expected Output

```
python
```

Test Case 2

Input values

```
2026AI
```

Expected Output

```
AI
```

10. pass Statement

Question 46: Print Even Numbers Using pass

Difficulty: Core

Read n. Visit every number from 1 to n. If the number is odd, execute pass; otherwise print the even number.

Input Format: One integer n.

Output Format: Print even numbers from 1 to n, one per line.

Practice Rule: The odd-number branch must contain pass.

Test Case 1

Input values

```
8
```

Expected Output

```
2
4
6
8
```

Test Case 2

Input values

```
5
```

Expected Output

```
2
4
```

Question 47: Long Words Only

Difficulty: Core

Read space-separated lowercase words. For each word shorter than 4 characters, execute pass. Otherwise, print the word.

Input Format: One line of space-separated words.

Output Format: Print words having at least 4 characters, one per line.

Practice Rule: Use if ... pass ... else.

Test Case 1

Input values

```
sun moon sky stars
```

Expected Output

```
moon
stars
```

Test Case 2

Input values

```
red blue green
```

Expected Output

```
blue
green
```

Question 48: Ignore Multiples of Five with pass

Difficulty: Core

Read n. Check numbers from 1 to n. If a number is divisible by 5, execute pass. Otherwise print the number.

Input Format: One integer n.

Output Format: Print all numbers not divisible by 5.

Practice Rule: Use pass in the divisible-by-5 branch and print in the else branch.

Test Case 1

Input values

7

Expected Output

1
2
3
4
6
7

Test Case 2

Input values

5

Expected Output

1
2
3
4

Question 49: Reserved Command Placeholder

Difficulty: Challenge

Read one command: START, PAUSE, or STOP. START prints Running, STOP prints Stopped, and PAUSE is intentionally not implemented yet. For PAUSE, execute pass and then print Feature Pending after the conditional block.

Input Format: One uppercase command.

Output Format: Print the message required by the command.

Practice Rule: The PAUSE branch must contain pass. Use a flag or another simple method so Feature Pending is printed only for PAUSE.

Test Case 1

Input values

START

Expected Output

Running

Test Case 2

Input values

PAUSE

Expected Output

Feature Pending

Question 50: Score Review Placeholder

Difficulty: Moderate

Read space-separated scores. For each score below 40, execute pass because the review action is not implemented. For every score 40 or above, print score: value.

Input Format: One line of integer scores.

Output Format: Print only scores that are 40 or above, one per line in the shown format.

Practice Rule: Use if ... pass ... else inside a for loop.

Test Case 1

Input values

35 70 22 49

Expected Output

```
score: 70
score: 49
```

Test Case 2

Input values

```
40 39 90
```

Expected Output

```
score: 40
score: 90
```

11. for-else Statement

Question 51: Prime Number Check

Difficulty: Core

Read an integer n greater than 1. Determine whether it is Prime or Not Prime. Stop the loop if a divisor is found; otherwise use the loop's else block to identify a prime number.

Input Format: One integer $n > 1$.

Output Format: Print Prime or Not Prime.

Practice Rule: Use for-else and break.

Test Case 1

Input values

```
13
```

Expected Output

```
Prime
```

Test Case 2

Input values

```
18
```

Expected Output

```
Not Prime
```

Question 52: Search for a Target

Difficulty: Core

Read a line of space-separated integers and then a target value. Search for the target. If found, print Found and stop. If the loop finishes without finding it, print Not Found.

Input Format: First line: integers. Second line: target integer.

Output Format: Print Found or Not Found.

Practice Rule: Use for-else.

Test Case 1

Input values

```
4 7 10 15
10
```

Expected Output

```
Found
```

Test Case 2

Input values

```
1 3 5 7
6
```

Expected Output

```
Not Found
```

Question 53: Validate a Numeric Code

Difficulty: Core

Read a string code. Check every character. If any character is not between 0 and 9, print Invalid and stop. If the loop completes normally, print Valid.

Input Format: One string code.

Output Format: Print Valid or Invalid.

Practice Rule: Use for-else. Do not use str.isdigit().

Test Case 1

Input values

```
508241
```

Expected Output

Valid

Test Case 2

Input values

31A90

Expected Output

Invalid

Question 54: Perfect Divisor Search

Difficulty: Moderate

Read n and a target divisor d . Search the proper divisors of n from 2 up to $n-1$. If d is encountered as a divisor, print Target divisor found and stop. If the loop finishes without finding d as a divisor, print Target divisor not found.

Input Format: Two integers n and d on separate lines.

Output Format: Print exactly one search result message.

Practice Rule: Use for-else and break only when both conditions are satisfied.

Test Case 1

Input values

24

6

Expected Output

Target divisor found

Test Case 2

Input values

24

5

Expected Output

Target divisor not found

Question 55: Username Validation

Difficulty: Challenge

Read a username. A valid username may contain lowercase letters, uppercase letters, and digits only. Scan each character. If any other character appears, print Invalid and stop; if the scan completes normally, print Valid.

Input Format: One username string.

Output Format: Print Valid or Invalid.

Practice Rule: Use for-else. Do not solve the entire validation with one built-in method such as isalnum().

Test Case 1

Input values

Student26

Expected Output

Valid

Test Case 2

Input values

student_26

Expected Output

Invalid

12. Number Logic: GCD, Composite Numbers and Divisors

Question 56: GCD of Two Numbers

Difficulty: Core

Read two non-zero integers and calculate their greatest common divisor using the Euclidean algorithm. If either input is zero, print value must be non zero.

Input Format: Two integers x and y.

Output Format: Print gcd: value, or the error message.

Practice Rule: Use a while loop; do not use math.gcd().

Test Case 1

Input values

50
50

Expected Output

gcd: 50

Test Case 2

Input values

48
18

Expected Output

gcd: 6

Question 57: Composite Registration Number

Difficulty: Core

Read a registration number. A composite number is greater than 3 and has at least one divisor other than 1 and itself. Print Eligible if the number is composite; otherwise print Not Eligible.

Input Format: One integer n.

Output Format: Print Eligible or Not Eligible.

Practice Rule: Use a loop to test divisibility and stop when a divisor is found.

Test Case 1

Input values

12

Expected Output

Eligible

Test Case 2

Input values

13

Expected Output

Not Eligible

Question 58: Count Divisors

Difficulty: Core

Read a positive integer n and count how many positive divisors it has, including 1 and n .

Input Format: One positive integer n .

Output Format: Print the divisor count.

Practice Rule: Use a loop; do not store all divisors in a list.

Test Case 1

Input values

12

Expected Output

6

Test Case 2

Input values

17

Expected Output

2

Question 59: LCM Using GCD

Difficulty: Challenge

Read two positive integers. First calculate their GCD using the Euclidean algorithm, then calculate the LCM using $(a*b)/gcd$.

Input Format: Two positive integers a and b on one line.

Output Format: Print gcd: value on the first line and lcm: value on the second line.

Practice Rule: Use a while loop for GCD. Do not use `math.gcd()` or `math.lcm()`.

Test Case 1

Input values

12 18

Expected Output

gcd: 6
lcm: 36

Test Case 2

Input values

7 5

Expected Output

gcd: 1
lcm: 35

Question 60: Prime Factors with Repetition

Difficulty: Challenge

Read an integer n greater than 1 and print its prime factors in increasing order, including repeated factors. Example: 12 has factors 2 2 3.

Input Format: One integer $n > 1$.

Output Format: Print the prime factors on one line separated by spaces.

Practice Rule: Use division and loops. Do not use external libraries.

Test Case 1

Input values

60

Expected Output

2 2 3 5

Test Case 2

Input values

49

Expected Output

7 7

13. Applied Conditional Problems

Question 61: Festival Bonus

Difficulty: Core

A company gives a 5% bonus to male employees and a 10% bonus to female employees. If salary is less than Rs. 10000, an additional 2% of salary is added to the bonus. Read salary and gender (M or F), then print the bonus and total salary, each formatted to two decimals.

Input Format: First line: floating-point salary. Second line: M or F.

Output Format: First line: bonus. Second line: salary plus bonus. Both to two decimals.

Practice Rule: Use conditional statements and percentage calculations.

Test Case 1

Input values

```
5000
M
```

Expected Output

```
350.00
5350.00
```

Test Case 2

Input values

```
12000
F
```

Expected Output

```
1200.00
13200.00
```

Question 62: Concert Ticket Cost

Difficulty: Core

There are N economy tickets, each costing c . Your queue position p is 1-based. If $p \leq N$, you buy an economy ticket for c . Otherwise, you buy a VIP ticket whose price is c plus 5 for every position beyond N . Print the amount you must pay.

Input Format: Three integers N , c , and p on separate lines.

Output Format: Print the integer ticket amount.

Practice Rule: Use a simple if-else decision.

Test Case 1

Input values

```
10
100
7
```

Expected Output

```
100
```

Test Case 2

Input values

```
10
100
14
```

Expected Output

```
120
```

Question 63: Type of Triangle

Difficulty: Core

Read three side lengths. Print Equilateral if all three are equal, Isosceles if exactly two sides are equal, and Scalene if all sides are different.

Input Format: Three integers a , b , and c .

Output Format: Print Equilateral, Isosceles, or Scalene.

Practice Rule: Use if-elif-else.

Test Case 1

Input values

```
5
5
5
```

Expected Output

Equilateral

Test Case 2

Input values

7
7
10

Expected Output

Isosceles

Question 64: ATM Withdrawal

Difficulty: Challenge

Read account balance and withdrawal amount. A withdrawal is allowed only if the amount is positive, is a multiple of 100, and does not exceed the balance. If allowed, print the new balance. Otherwise print Invalid amount, Insufficient balance, or Amount must be multiple of 100 as appropriate.

Input Format: Read balance and withdrawal on one line as two integers.

Output Format: Print exactly one result message or new balance: value.

Practice Rule: Use ordered conditional checks so that one clear message is produced.

Test Case 1

Input values

5000 1200

Expected Output

new balance: 3800

Test Case 2

Input values

5000 1250

Expected Output

Amount must be multiple of 100

Question 65: Delivery Charge Calculator

Difficulty: Challenge

Read order amount and delivery zone. Zones are LOCAL, CITY, and OUTSIDE. Delivery charge is 30, 60, and 120 respectively. If order amount is at least 1000, LOCAL and CITY delivery become free, but OUTSIDE still costs 120. Invalid zones print Invalid zone.

Input Format: Use `amount = float(input("Order amount: "))` and `zone = input("Zone: ")`.

Output Format: Print `charge:` followed by the charge formatted to two decimals, or Invalid zone.

Practice Rule: Use conditions and prompted input. Be careful that the free-delivery rule does not apply to OUTSIDE.

Test Case 1

Input values

```
800  
CITY
```

Expected Output

```
charge: 60.00
```

Test Case 2

Input values

```
1500  
LOCAL
```

Expected Output

```
charge: 0.00
```

Final Practice Checklist

By the end of this workbook, you should be able to solve problems involving:

- plain input, prompted input, one-line multi-value input with `split()`, output formatting, arithmetic, `//` and `%`;
- `if`, `elif`, `else`, nested decisions, cumulative slabs, validation, and multi-condition rules;
- match-case with values, strings, guards, and menu-style commands;
- `while` loops, counters, digit processing, repeated attempts, and condition-controlled repetition;
- `for` loops, character/list traversal, counters, accumulators, and manual second-largest logic;
- `range()` with start, stop, positive steps, and negative steps;
- nested loops for patterns and multiplication grids;
- `break`, `continue`, `pass`, sentinels, filtering, and early termination;
- `for-else` search and validation logic;
- GCD, LCM, divisors, composite numbers, prime factors, and other number logic; and
- applied decision problems involving billing, authentication, ATM rules, salary, tickets, and delivery charges.

Self-challenge: For at least 10 questions, change the input values, test boundary cases, and predict the output before running your program.

Build the logic first. Then write the syntax. Then test the boundaries.